# DSA06 DATA HANDLING AND VISUALIZATION

Activity: Correlation Detective | Slot D | Case File CD-D12: The Campus Library Case

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## Question Paper

**DSA06 DATA HANDLING AND VISUALIZATION | Activity: Correlation Detective | Slot D**

**R.No: 12 Reg. No: 192424277 Name: GUNA P**

*Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; your case file is different from your neighbour's. Write your answers and paste your plots and code.*

### Case File CD-D12: The Campus library Case [20 marks]

**Briefing:** You are the data detective hired by the head librarian of a university campus. The file holds 17 daily records from the exam month, with seven variables. Some relationships can be trusted. Others are traps: a hidden driver, a curve that fools r, and a single wrong record. Find them.

| R name | What it measures | R name | What it measures |
|---|---|---|---|
| pressure | Exam-pressure index (0-100) | study_complaints | Study-room complaints |
| coffee | Coffee cups sold | group_sessions | Group sessions booked |
| late_entries | Late-night entries | print_jobs | Print jobs |
| room_dev | Room temperature minus 22 deg C | | |

**Evidence: run this code in R to create your case file (do not edit it).**

```
set.seed(21651)
n <- 17
pressure <- round(rnorm(n, 52.1, 11.0), 1)
coffee <- round(156 + 41.4 * (0.92 * (pressure - 52.1) / 11.0 + 0.38 * rnorm(n)), 0)
late_entries <- round(61 + 16.6 * (-0.88 * (pressure - 52.1) / 11.0 + 0.47 * rnorm(n)), 0)
room_dev <- round(seq(-4.1, 4.1, length.out = n)[order(runif(n))], 1)
study_complaints <- round(2.6 + 13.6 * (room_dev / 4.1)^2 + rnorm(n, 0, 0.4), 0)
group_sessions <- round(rnorm(n, 10, 2.8), 0)
print_jobs <- round(208 + 55.2 * (0.10 * (group_sessions - 10) / 2.8 + 0.99 * rnorm(n)), 0)
group_sessions[12] <- 20; print_jobs[12] <- 380   # row 12 was typed in by a trainee
case <- data.frame(pressure, coffee, late_entries, room_dev, study_complaints, group_sessions, print_jobs)
```

**Tasks:**

1. The matrix. Use cor(case) to get the 7 x 7 correlation matrix (round to 2 decimals). State n and how many distinct pairs of variables there are. Draw a correlogram of the matrix (ggplot2 geom_tile with geom_text, or corrplot). [4]
2. The strongest clues. Name the strongest positive pair and the strongest negative pair (off-diagonal). Give r for each and call it weak (below 0.3), moderate (0.3 to 0.7) or strong (above 0.7). [3]
3. The deceptive zero. Plot study_complaints against room_dev with a straight line, then add a quadratic fit (formula = y ~ x + I(x^2)). Report r and the R-squared of the quadratic fit. In one sentence, say why r misses this relationship. [4]
4. The outlier alibi. Plot print_jobs against group_sessions. Find the row that looks wrong and give its row number. Report r for all rows and r without that row, and say what the single record did to r. [4]
5. The prime suspect. coffee and late_entries are strongly linked. Report r(coffee, late_entries), then r(pressure, coffee) and r(pressure, late_entries). Compute the partial correlation of coffee and late_entries controlling for pressure: correlate resid(lm(coffee ~ pressure)) with resid(lm(late_entries ~ pressure)). Write a two-sentence verdict: does coffee cause late_entries? Name one study design that could test causation. [5]

## Case File Creation (given code, not edited)

```
set.seed(21651)
n <- 17
pressure <- round(rnorm(n, 52.1, 11.0), 1)
coffee <- round(156 + 41.4 * (0.92 * (pressure - 52.1) / 11.0 + 0.38 * rnorm(n)), 0)
late_entries <- round(61 + 16.6 * (-0.88 * (pressure - 52.1) / 11.0 + 0.47 * rnorm(n)), 0)
room_dev <- round(seq(-4.1, 4.1, length.out = n)[order(runif(n))], 1)
study_complaints <- round(2.6 + 13.6 * (room_dev / 4.1)^2 + rnorm(n, 0, 0.4), 0)
group_sessions <- round(rnorm(n, 10, 2.8), 0)
print_jobs <- round(208 + 55.2 * (0.10 * (group_sessions - 10) / 2.8 + 0.99 * rnorm(n)), 0)
group_sessions[12] <- 20; print_jobs[12] <- 380
case <- data.frame(pressure, coffee, late_entries, room_dev, study_complaints, group_sessions, print_jobs)
```

### Output: `print(case)` (17 rows x 7 variables)

| | pressure | coffee | late_entries | room_dev | study_complaints | group_sessions | print_jobs |
|---|---|---|---|---|---|---|---|
| **1** | 45.5 | 126 | 64 | -3.6 | 13 | 11 | 222 |
| **2** | 54.8 | 189 | 64 | 0.5 | 2 | 9 | 270 |
| **3** | 45.3 | 154 | 62 | 1.5 | 5 | 10 | 253 |
| **4** | 42.6 | 119 | 78 | 2.6 | 8 | 11 | 177 |
| **5** | 36.7 | 99 | 87 | -1.5 | 5 | 6 | 83 |
| **6** | 58.5 | 197 | 47 | -2.6 | 9 | 7 | 157 |
| **7** | 27.5 | 66 | 101 | -0.5 | 3 | 6 | 258 |
| **8** | 57.8 | 168 | 66 | 0.0 | 3 | 6 | 122 |
| **9** | 53.1 | 116 | 58 | 3.1 | 10 | 9 | 139 |
| **10** | 41.2 | 96 | 77 | 4.1 | 17 | 9 | 189 |
| **11** | 49.6 | 143 | 67 | -2.0 | 6 | 10 | 160 |
| **12** | 53.9 | 177 | 56 | 2.0 | 6 | 20 | 380 |
| **13** | 51.3 | 159 | 69 | -3.1 | 10 | 8 | 165 |
| **14** | 58.0 | 163 | 56 | -4.1 | 16 | 12 | 122 |
| **15** | 52.3 | 149 | 59 | 1.0 | 4 | 10 | 261 |
| **16** | 48.1 | 146 | 69 | -1.0 | 3 | 11 | 240 |
| **17** | 54.6 | 156 | 63 | 3.6 | 13 | 8 | 246 |

# Task 1. The matrix [4 marks]

## R code

```
cm <- round(cor(case), 2)
cm
n <- nrow(case)
n
choose(ncol(case), 2)

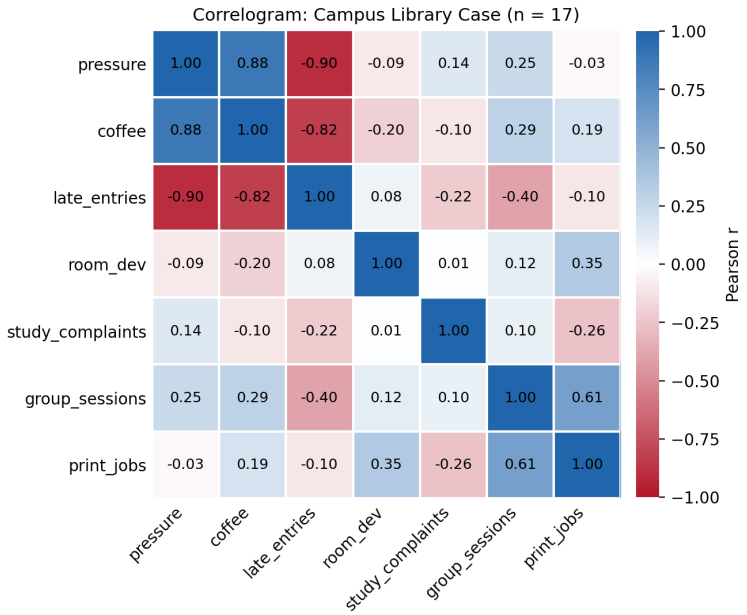
library(ggplot2)
cm_long <- as.data.frame(as.table(cm))
ggplot(cm_long, aes(Var1, Var2, fill = Freq)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = sprintf("%.2f", Freq)), size = 3.5) +
  scale_fill_gradient2(low = "#B2182B", mid = "white", high = "#2166AC",
                       midpoint = 0, limits = c(-1, 1), name = "Pearson r") +
  coord_fixed() +
  labs(title = "Correlogram: Campus Library Case (n = 17)", x = NULL, y = NULL) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

## Output: correlation matrix (`round(cor(case), 2)`)

| | pressure | coffee | late_entries | room_dev | study_complaints | group_sessions | print_jobs |
|---|---|---|---|---|---|---|---|
| **pressure** | 1.00 | 0.88 | -0.90 | -0.09 | 0.14 | 0.25 | -0.03 |
| **coffee** | 0.88 | 1.00 | -0.82 | -0.20 | -0.10 | 0.29 | 0.19 |
| **late_entries** | -0.90 | -0.82 | 1.00 | 0.08 | -0.22 | -0.40 | -0.10 |
| **room_dev** | -0.09 | -0.20 | 0.08 | 1.00 | 0.01 | 0.12 | 0.35 |
| **study_complaints** | 0.14 | -0.10 | -0.22 | 0.01 | 1.00 | 0.10 | -0.26 |
| **group_sessions** | 0.25 | 0.29 | -0.40 | 0.12 | 0.10 | 1.00 | 0.61 |
| **print_jobs** | -0.03 | 0.19 | -0.10 | 0.35 | -0.26 | 0.61 | 1.00 |

`[1] 17` (n)  `[1] 21` (choose(7, 2))

## Output: correlogram




**Answer:**

Number of records n = 17 (one for each day). There are 7 variables, so the number of different pairs is 7 x 6 / 2 = 21. In the correlogram, pressure, coffee and late_entries have dark cells (strong correlation). Most other cells are light, which means weak correlation. The only other noticeable one is group_sessions and print_jobs (0.61).

## Task 2. The strongest clues [3 marks]

### R code

```
idx <- which(upper.tri(cm), arr.ind = TRUE)
pairs <- data.frame(var1 = rownames(cm)[idx[, 1]],
                    var2 = colnames(cm)[idx[, 2]],
                    r = cm[idx])
pairs[order(pairs$r), ]
pairs[which.max(pairs$r), ]
pairs[which.min(pairs$r), ]
```

### Output

Three most negative and three most positive of the 21 pairs:

| var1 | var2 | r |
|---|---|---|
| **pressure** | late_entries | -0.90 |
| **coffee** | late_entries | -0.82 |
| **late_entries** | group_sessions | -0.40 |
| **room_dev** | print_jobs | 0.35 |
| **group_sessions** | print_jobs | 0.61 |
| **pressure** | coffee | 0.88 |

`which.max` gives pressure - coffee, r = 0.88; `which.min` gives pressure - late_entries, r = -0.90.

**Answer:**
Strongest positive pair: pressure and coffee, r = 0.88 (strong).
Strongest negative pair: pressure and late_entries, r = -0.90 (strong).
coffee and late_entries (r = -0.82) is also strong. All three pairs involve pressure.

## Task 3. The deceptive zero [4 marks]

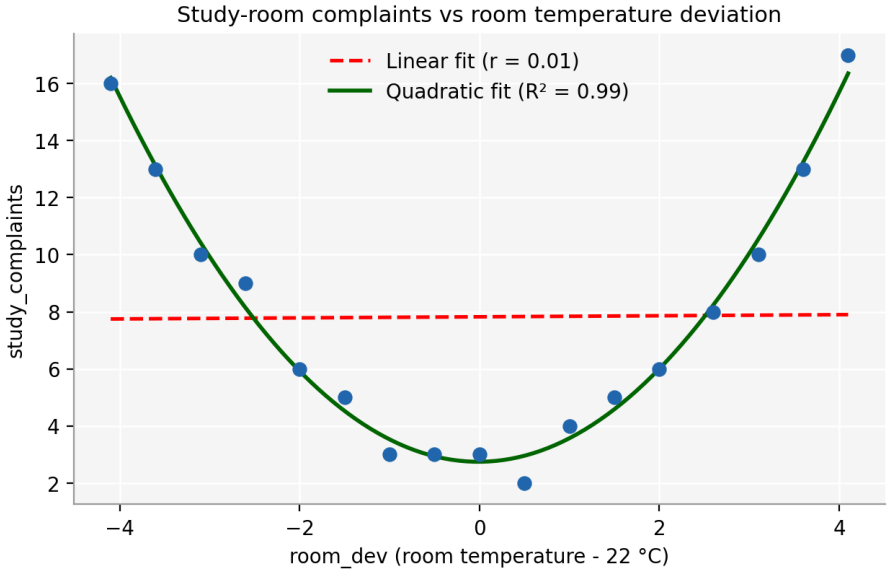
### R code

```
r3 <- cor(case$room_dev, case$study_complaints)
round(r3, 2)
fit <- lm(study_complaints ~ room_dev + I(room_dev^2), data = case)
round(summary(fit)$r.squared, 2)

ggplot(case, aes(room_dev, study_complaints)) +
  geom_point(colour = "#2166AC", size = 2.5) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE,
              colour = "red", linetype = "dashed") +
  geom_smooth(method = "lm", formula = y ~ x + I(x^2), se = FALSE,
              colour = "darkgreen") +
  labs(title = "Study-room complaints vs room temperature deviation",
       x = "room_dev (room temperature - 22 deg C)", y = "study_complaints") +
  theme_gray()
```

### Output

`[1] 0.01` (r) `[1] 0.99` (R-squared of quadratic)




**Answer:**

r = 0.01 and R-squared of the quadratic fit = 0.99.

r is almost zero because the relationship is U-shaped, not a straight line. r only measures straight-line relationship, so the left side (going down) and right side (going up) of the curve cancel each other, even though room_dev explains about 99% of study_complaints.

## Task 4. The outlier alibi [4 marks]

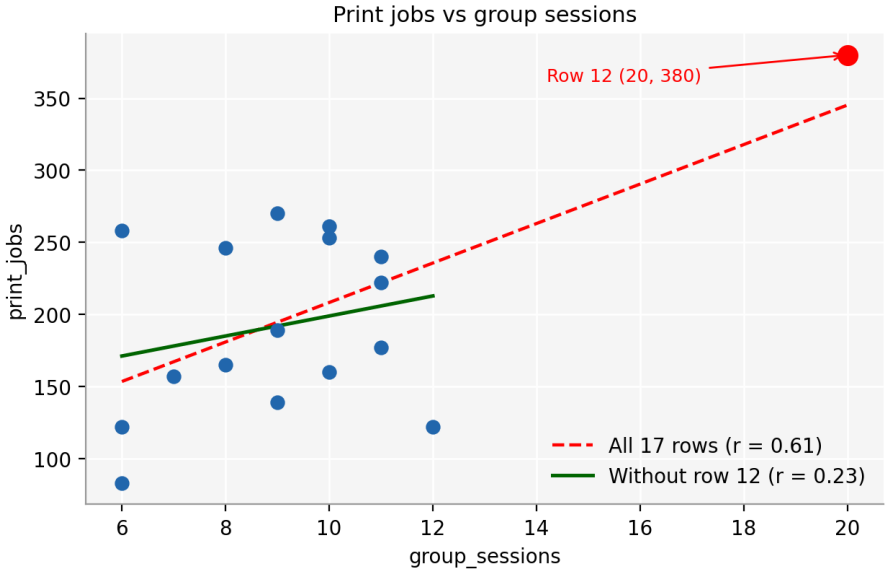
### R code

```
cor(case$group_sessions, case$print_jobs)
which.max(case$print_jobs)
case[12, ]
cor(case$group_sessions[-12], case$print_jobs[-12])

ggplot(case, aes(group_sessions, print_jobs)) +
  geom_point(colour = "#2166AC", size = 2.5) +
  geom_point(data = case[12, ], colour = "red", size = 4) +
  geom_smooth(method = "lm", se = FALSE, colour = "red", linetype = "dashed") +
  geom_smooth(data = case[-12, ], method = "lm", se = FALSE, colour = "darkgreen") +
  annotate("text", x = 17, y = 370, label = "Row 12 (20, 380)", colour = "red") +
  labs(title = "Print jobs vs group sessions") +
  theme_gray()
```

### Output

`[1] 0.61` (r, all 17 rows) `[1] 12` (suspect row) `[1] 0.23` (r without row 12)




**Answer:**
The wrong record is row 12 (group_sessions = 20, print_jobs = 380). It is far away from all the other points (the next highest values are 12 sessions and 270 print jobs).
r for all rows = 0.61 (moderate). r without row 12 = 0.23 (weak).
This one wrongly typed record pulled the line upwards and increased r from 0.23 to 0.61, so it created a correlation that is not really there.

## Task 5. The prime suspect [5 marks]

### R code

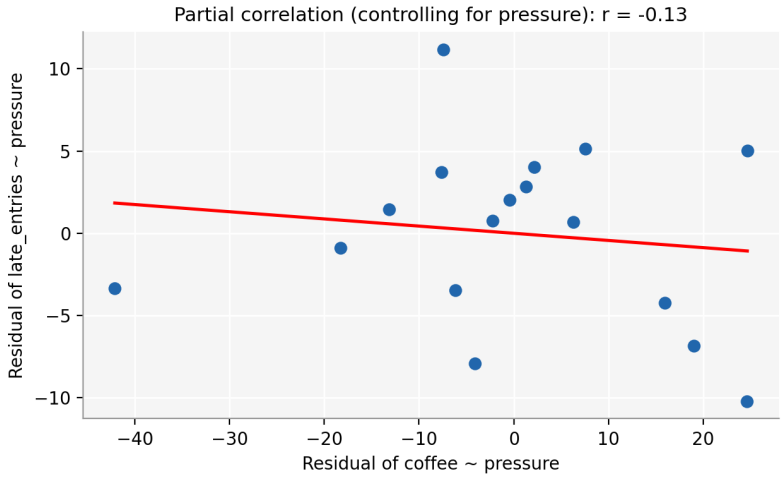
```
r_cl <- cor(case$coffee, case$late_entries)
r_pc <- cor(case$pressure, case$coffee)
r_pl <- cor(case$pressure, case$late_entries)
round(c(r_cl, r_pc, r_pl), 2)

res_coffee <- resid(lm(coffee ~ pressure, data = case))
res_late   <- resid(lm(late_entries ~ pressure, data = case))
round(cor(res_coffee, res_late), 2)

ggplot(data.frame(res_coffee, res_late), aes(res_coffee, res_late)) +
  geom_point(colour = "#2166AC", size = 2.5) +
  geom_smooth(method = "lm", se = FALSE, colour = "red") +
  labs(title = "Partial correlation (controlling for pressure)",
       x = "Residual of coffee ~ pressure", y = "Residual of late_entries ~ pressure") +
  theme_gray()
```

### Output

| r(coffee, late_entries) | r(pressure, coffee) | r(pressure, late_entries) | Partial r (coffee, late_entries \| pressure) |
|---|---|---|---|
| -0.82 | 0.88 | -0.90 | -0.13 |




**Answer:**
r(coffee, late_entries) = -0.82, r(pressure, coffee) = 0.88, r(pressure, late_entries) = -0.90.
Partial correlation of coffee and late_entries controlling for pressure = -0.13 (weak).
**Verdict:** No, coffee does not cause late_entries. The correlation of -0.82 drops to only -0.13 once pressure is kept constant, so exam pressure is affecting both coffee and late_entries. Correlation alone does not prove causation.
**Study design:** A randomised controlled trial - randomly give students coffee or decaf (without telling them) during exam weeks with similar pressure, and then compare the number of late_entries in the two groups.